Highlights

Soil-geomorphic controls on berm structural condition and vegetation response in the Altar Valley, southern Arizona

- Legacy water spreader berms in the Altar Valley, southern Arizona, vary widely in structural condition and vegetation impact
- Remote sensing and soil survey data elucidate soil-geomorphic controls on berm condition and vegetation response across 775 berms in the Altar Valley, southern Arizona
- Berm length, flow accumulation, and soil texture control structural condition, while slope and soil texture control vegetation response; the two outcomes are statistically independent

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Abstract

Earthen berms are commonly constructed on rangelands in the southwestern US to control water runoff, increase soil moisture, and reduce soil erosion. Many legacy berms date to the 1930s, and their condition and ecohydrologic impacts are poorly understood. Here we analyze 775 water spreader berms across the Altar Valley, southern Arizona, to determine how soil properties, flow accumulation, berm length, and slope influence structural condition and vegetation response. Berms are more likely to remain intact when they are shorter, located in positions of lower flow accumulation, and built on finer-textured soils, whereas vegetation responses are larger on steeper slopes and coarser-textured (sandy loam) soils. Nearly half of berms (47%) exceed a 7% SAVI-difference threshold indicative of measurable vegetation response. Structural condition and vegetation response are statistically independent ($\varphi \approx 0$), indicating that intact berms do not necessarily produce stronger vegetation responses. Landform is univariately associated with vegetation response but loses significance after controlling for slope and soil properties, indicating confounding effects of other variables. These findings demonstrate the importance of incorporating site-specific landscape features when planning and managing berms for erosion control and vegetation restoration in semi-arid regions.

Keywords: semi-arid rangelands, water spreader berms, soil-geomorphic controls, berm structural condition, vegetation response, Altar Valley

Plain language summary

Starting in the 1930s, thousands of earthworks, such as water spreader berms, were constructed across southwestern United States rangelands to help control water runoff and prevent soil erosion. Water spreader berms function by slowing down rainwater, allowing it to infiltrate into the soil, which increases soil moisture and supports vegetation growth. However, over time, berms can degrade, potentially reducing their effectiveness [1]. We examined how soil type, slope, landform, and berm

length affect both berm integrity and vegetation response nearby. We quantified berm impacts on surrounding vegetation for 775 berms in the Altar Valley, Arizona, using the Soil Adjusted Vegetation Index (SAVI), a satellite-based measure of vegetation greenness similar to NDVI. We found that berm structural condition – intact or degraded – is determined primarily by berm length, flow accumulation, and soil texture: shorter berms on finer-textured soils with lower flow accumulation are more likely to remain intact. Vegetation response – how much greener upslope than downslope – is better predicted by slope and soil texture: berms on steeper slopes and coarser-textured (sandy loam) soils exhibit larger differences in vegetation greenness upslope vs. downslope. Whether a berm is intact or degraded does not predict vegetation response, suggesting that these two outcomes are shaped by different environmental controls. This study demonstrates that considering the specific characteristics of the landscape can help land managers make better decisions about where to place berms in rangeland landscapes, and inform decision making about berm maintenance and removal.

1. Introduction

As commercial cattle ranching spread through the southwestern United States in the late 1800s [?], widespread soil loss and land degradation prompted conservation interventions. Beginning in the 1930s, the US Soil Conservation Service (now the USDA Natural Resources Conservation Service, NCRS) partnered with private landowners to fund and construct water management structures, aiming to distribute cattle, maintain forage, and mitigate land degradation [?]. These collaborations resulted in thousands of earthen berms, stock tanks, water spreader berms, and other conservation structures across the region. The legacy of these structures continues to shape hydrological and ecological patterns in the US Southwest [? ?]. Water spreader berms, in particular, were typically constructed as large earthen berms (often more than 100 m long and 3–5 m wide) along landscape contours, either singly or in series [?]. Berms constructed perpendicular to the flow direction reduce flow velocities, allowing time for runoff to infiltrate and increase soil moisture, which is critical for supporting vegetation [?]. Earthen berms were also constructed to capture sediment and reduce gully erosion, for example, constructed upslope of a gully to halt head cutting. Because of their influence on water distribution and sediment movement, earthworks were often integral components of efforts to restore degraded rangelands. Earthen berms are subject to failure over time. Many rangeland berms have been breached (a break through a berm), or flanked (concentrated runoff around the end of a berm) [1]. Without maintenance, their deterioration can impact water and sediment flows, and influence vegetation and soil distribution across rangeland landscapes [1]. Early assessments of rangeland conservation structures dating to the 1930s indicated high rates of structural failure with limited vegetation response [?]. Recent research has focused on the impacts of rangeland berms on geomorphic evolution [?] and the ecological impact of berms on vegetation patterns [?]. However, few studies have addressed how landscape controls — such as geomorphic position, soils, and local topography — influence berm condition and vegetation response. Likewise, whether berm failure impacts vegetation response is also an unknown. Here, we examine how the landscape setting shapes the structural condition of water spreader berms and their impact on surrounding vegetation. We use controlled logistic regression to isolate the independent contributions of landform, soil texture, soil development, berm length, surface slope, and flow accumulation to berm structural condition (intact vs. degraded) and vegetation response, complemented by a Random Forest classifier to assess overall predictive performance. Vegetation response is characterized using the Soil Adjusted Vegetation Index (SAVI; [?]), a vegetation index adjusted for bright soils typical of desert environments. Following [?], we quantify vegetation response as the percent difference in SAVI between areas immediately upslope and downslope of berms, within 60 m, normalized by background conditions (ΔS). Predicting berm structural condition and vegetation response from site-specific physical properties can inform decisions about berm placement, maintenance, and removal priorities.

2. Methods

2.1. Site description: Altar Valley

This study focuses on water spreader berms in the Altar Valley, a semi-arid, 247,000 ha watershed in southern Arizona, USA, where more than 1,500 manmade structures have been constructed since the early 1900s to alter surface runoff [1]. Runoff in the valley is generated primarily in response to intense and infrequent summer convective thunderstorms, with more than half of the 415 mm of annual rainfall occurring from July to September during the North American Monsoon [?]. We use an inventory of 775 mapped water spreader berm locations [1] (see Figure 1). This dataset classifies berms as intact, breached, or flanked using a combination of Google Earth Imagery and aerial LiDAR data (2011 and 2016; Pima County Regional Flood Control District) [2], and includes berm type, structural condition, and berm length.

2.2. Data

Topography

Two 1/3 arc-second USGS National Map DEM GeoTIFF tiles were downloaded from the USGS National Map Downloader and processed in ArcGIS Pro 3.3.1. The tiles were mosaicked into a single elevation raster and clipped to the dissolved boundary of the Brawley Wash and Río de la Concepción HUC8 watersheds. To generate flow accumulation, the clipped DEM was hydrologically conditioned using the Fill tool to remove sinks. Flow direction was then computed from the filled DEM using the D8 algorithm. Finally, Flow Accumulation was calculated from the D8 flow-direction raster and exported as a GeoTIFF.

NRCS soils data

Landform, soil texture, and soil development (B-horizon presence) at each berm location were obtained from USDA Natural Resources Conservation Service (NRCS) soil survey data. Landform influences water flow, erosion potential, and sediment deposition, all of which may impact berm stability and surrounding vegetation. Soil texture – the relative proportions of sand, silt, and clay – influences soil drainage capacity, cohesion, and susceptibility to erosion. The presence or absence of a B horizon (Bt or Bk) – defined as a subsurface horizon of accumulated illuvial clay (Bt) or pedogenic carbonate (Bk) – reflects the degree of soil maturity and horizon differentiation, including processes such as clay translocation and carbonate precipitation [? ?]. Soil texture and development both influence soil structural stability and water retention, and likely impact berm integrity over time. Landform in the Altar Valley falls into three main categories: flood plains, stream terraces, and fan terraces. Soil texture in the Altar Valley falls into 8 NRCS classes; analyses focus on the three most common: clay loam, sandy loam, and silt loam (Table 1). Soil development was classified as B horizon present (Bt or Bk horizon, indicating illuvial clay or carbonate accumulation) vs. absent (A-C profile). Berm length and slope were each split at their respective sample medians (60 m and 2%), yielding approximately balanced groups for univariate analyses (Table 1). Soil texture and

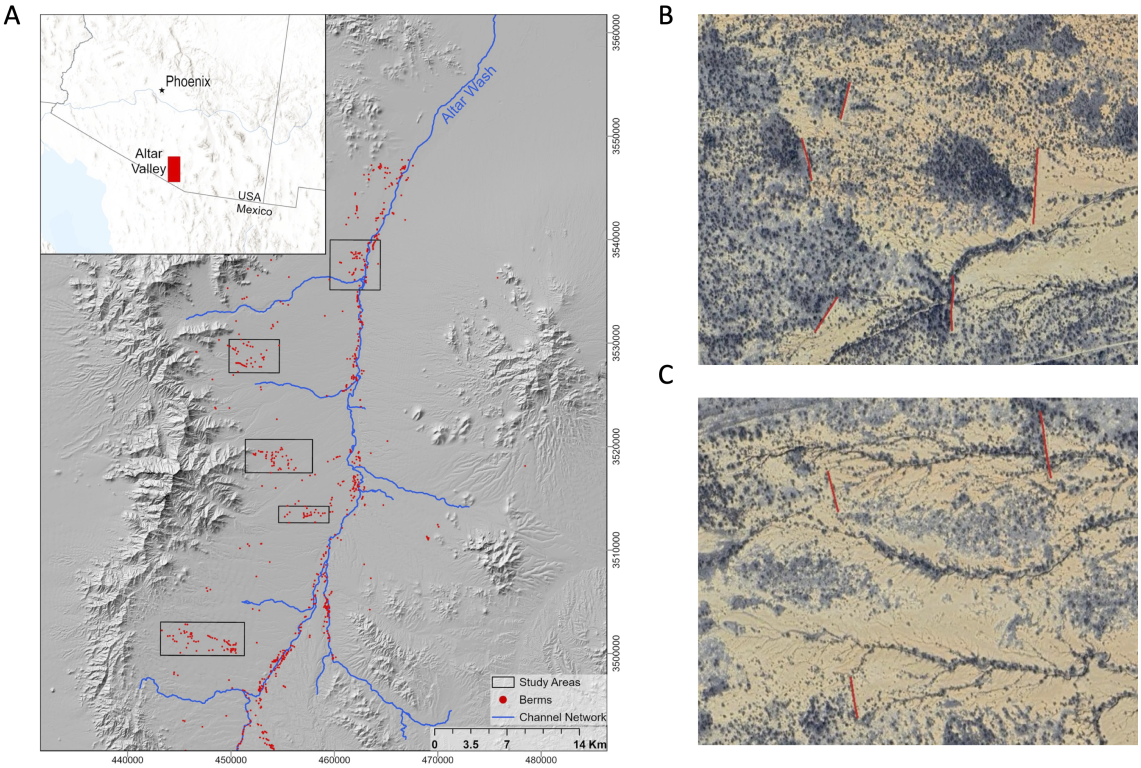

Figure 1: Study area map showing berm locations across three landform types in the Altar Valley, southern Arizona. Fan terraces and flood plains contain the majority of berms, with stream terraces forming a narrower transitional band.

development are included as predictors because both have been linked to earthwork stability and dryland vegetation response [? ? ? ?]; mechanisms and supporting literature are discussed in Section 4.6.

2.3. Vegetation response

Sentinel-2 data products (10 m resolution; 2016–2024; [3]) were used to compute the Soil Adjusted Vegetation Index (SAVI) and derive metrics of berm impact on surrounding vegetation. SAVI S was computed as:

$$S = \frac{(NIR - RED)}{(NIR + RED + L)} \times (1 + L) \tag{1}$$

where NIR is the near-infrared reflectance, RED is the red reflectance, and L is the soil brightness correction factor, typically set to 0.5 [?]. Following ?], we quantified the difference in SAVI between areas 15–60 m upslope and 15–60 m downslope of each berm, normalized by background values from areas within 1000 m but more than 200 m from any berm (Figure 2A,B). Berm impacts on vegetation

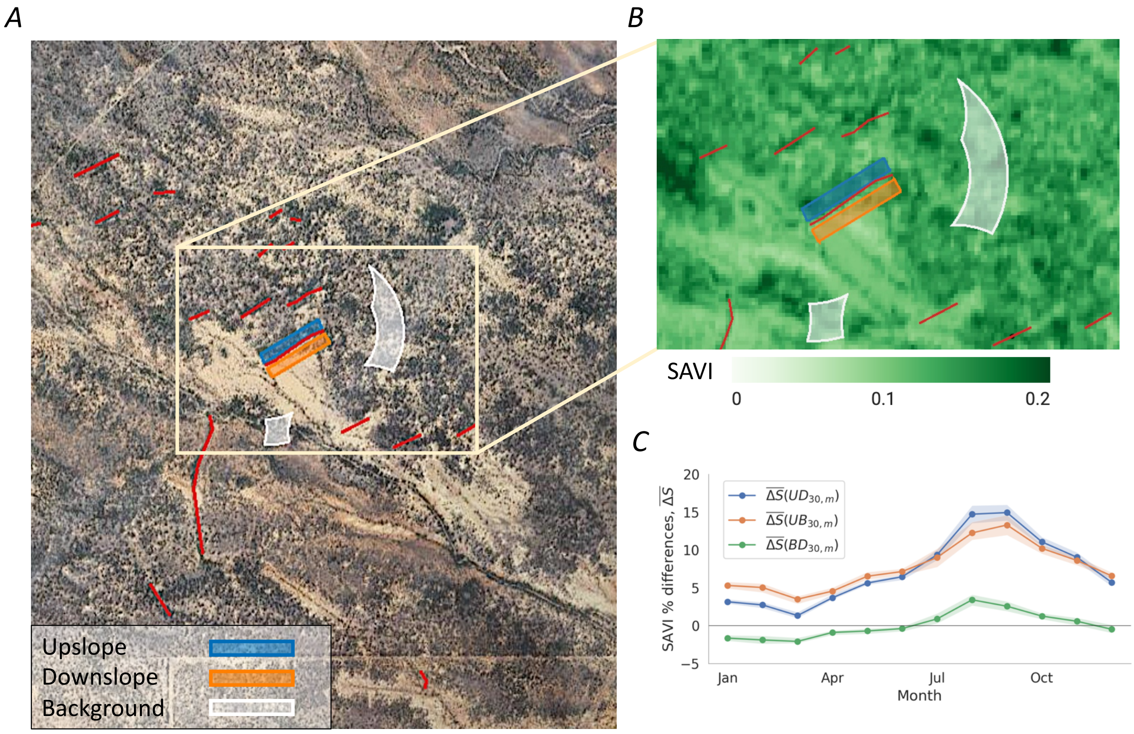

Figure 2: Vegetation response metric for an example berm. (A) Upslope, downslope, and background area delineation. (B) SAVI values in these three areas. (C) Monthly climatology of SAVI showing the upslope–downslope contrast used to compute ΔS.

are strongest at the end of the monsoon season (August–September; Figure 2C): **?**] showed that berms enhance upslope SAVI by ~15% relative to background conditions in August–September, with effects concentrated to within 60 m of the berms. Cohen's d effect sizes for the upslope–downslope SAVI difference at $L = 30$ m (15–45 m from the berm) were large ($d > 0.8$) during monsoon months and negligible ($d < 0.2$) in winter, with effect sizes decreasing with increasing distance. To quantify local berm impacts on vegetation greenness, we calculated the percent SAVI difference between upslope and downslope areas, ΔS:

$$\Delta S = \frac{S_U - S_D}{S_B} \times 100 \tag{2}$$

where S_U, S_D, and S_B are the median SAVI values in the upslope, downslope, and background areas, respectively. For each berm, median SAVI was computed over Augusts and Septembers across 2016–2024 within 15–60 m upslope and downslope of the berm. The background value S_B was computed over the same months from areas of equivalent landform and soil type located beyond

Table 1: Variable categories and corresponding number of berms (N). Breached and flanked berms are combined as "Degraded" for binary analyses.

Variable	**Category**	**N**
Response variables		
Berm structural condition	Intact	458
	Degraded (breach + flank)	317
Vegetation response	Vegetation response	368
	No vegetation response	407
Predictor variables		
Landform	Flood plains	359
	Fan terraces	276
	Stream terraces	140
Soil texture	Clay loam	283
	Sandy loam	210
	Silt loam	137
	Other[a]	145
B-horizon presence	B horizon	289
	No B horizon	486
Berm length	≤ 60 m	399
	> 60 m	376
Slope	$\leq 2\%$	410
	$> 2\%$	365
Flow accumulation	Continuous (contributing cells)	

[a]Other includes fine sandy loam ($n = 54$), loam ($n = 52$), sandy clay loam ($n = 22$), loamy coarse sand ($n = 9$), and loamy sand ($n = 8$).

the influence of any berm (>200 m from the nearest structure). We express the difference as a percentage because absolute SAVI values in the Altar Valley are small, typically 0.05–0.2 [?]. For the analyses presented here, the continuous ΔS metric was converted to a binary outcome: berms with $\Delta S > 7\%$ were classified as exhibiting a *vegetation response*, while those with $\Delta S \leq 7\%$ were classified as *no vegetation response*. The 7% threshold was informed by the Cohen's d sensitivity analysis in [?], which showed that SAVI differences below $\sim$7% correspond to small or negligible effect sizes ($d < 0.5$), whereas differences above this level are associated with medium to large effect sizes. This threshold is used to distinguish SAVI differences attributable to background variability from SAVI differences that indicate a vegetation response to berm presence. Sensitivity analysis indicated that the results do not change qualitatively if this threshold is varied. Of the 775 berms in this analysis, 368 (47%) exceeded the $\Delta S = 7\%$ threshold.

Statistical approach

We first assessed univariate associations between each predictor variable and each outcome (intact vs. degraded; vegetation response vs. no response). For binary predictors (e.g., B-horizon presence), we used two-sided Fisher's exact tests, which are preferable to chi-square tests in cases with small or unbalanced count data [?]. For multi-category predictors (e.g., landform and soil texture), we first tested for overall association with chi-square tests of independence; where the overall test was significant we then conducted pairwise two-sided Fisher's exact tests across all category pairs, adjusting the resulting p-values with the Benjamini–Hochberg correction to control the false discovery rate (the expected proportion of false positives). We used the Benjamini–Hochberg rather than Bonferroni correction because Bonferroni becomes overly stringent as the number of pairwise comparisons grows, sharply reducing power to detect real differences [?]. For univariate analysis, we binarized continuous variables near the sample medians: berm length at 60 m (median 58.5 m), slope at 2% (median 1.8%), and flow accumulation at 2,000 contributing 30 m cells (median 2,054).

Interpretation of the univariate analysis is complicated by the fact that the predictor variables are highly correlated — for example, fan terraces generally have more steeply sloping terrain than flood plains. We therefore conducted separate analyses to isolate the independent contributions of the predictor variables to berm condition and vegetation response. For each of these response variables, we fit a binomial logistic regression that included all predictors and assessed each via drop-one Likelihood Ratio Tests (LRT) [?].

For each predictor, we compared the full model to a reduced model with that predictor removed. The resulting ΔAIC quantifies the information lost by omitting a predictor ($\Delta\text{AIC} > 2$ indicates a meaningful independent contribution). The controlled p-value indicates the predictor's significance with the remaining predictors constant, and comparing univariate and controlled p-values distinguishes confounding (a predictor loses significance when others are included) from suppression (a predictor gains significance once confounders are removed). As a complementary, model-agnostic check on the controlled regression results, we trained a Random Forest classifier on the same six predictors (berm length, flow accumulation, soil texture, B-horizon presence, slope, landform) for each outcome. Model performance was evaluated using 5-fold stratified cross-validation (reported as CV AUC), and predictor importance was assessed by permutation importance on out-of-fold predictions.

3. Results

3.1. Predictor structure

The candidate predictor variables were highly correlated (Figure 3): landform, soil texture, and soil development are strongly inter-associated (Bergsma–Wicher Cramér's $V = 0.89$–0.97), while berm structural condition and vegetation response are only weakly associated with any predictor or with each other. Principal components analysis (PCA) of the predictor space (Figure 4) shows that PC1 (38%) describes soil texture and slope, while PC2 (30%) captures landform and soil texture variation. In Figure 4C, landform clusters separate cleanly rather than along an independent landform dimension. This covariance among predictors motivates the controlled analyses below, which determines each predictor's independent contribution to berm condition and vegetation response.

3.2. Controls on berm structural condition

Of 775 berms, 458 (59%) were classified as intact and 317 (41%) as degraded (203 flanked, 114 breached). Univariate analyses (Figure 5) showed that berm length was the strongest univariate predictor of condition: long berms (>60 m) had significantly higher failure rates than short berms ($\leq$60 m; $p < 0.001$).

Berms in areas of high flow accumulation were likely to be degraded (flow accumulation $p < 0.001$). Among soil texture classes with $n > 100$, texture was highly significant (χ^2 $p < 0.001$): sandy loam berms were more frequently degraded, and clay loam and silt loam berms have comparable intactness rates (both ∼62%). Berms on soils containing a B horizon were more frequently degraded, although the univariate association fell short of significance ($p = 0.059$); this pattern is interpreted in Section 4.4. Landform was not significantly associated with condition (χ^2 $p = 0.204$; Figure 5). Hillslope gradient shows a weak univariate signal ($p = 0.016$); the 2% threshold separating shallow and steep classes is modest, and most berms occupy gentle terrain (median ∼2%). A sensitivity analysis using a steeper 5% threshold strengthens the slope signal: berms on slopes >5% are more frequently degraded, consistent with the expectation that slope-related mechanical failure becomes important once gradients exceed the gentle-terrain regime that dominates this dataset.

Controlled logistic regression analysis using drop-one likelihood ratio tests (LRT) quantified the independent contribution of each predictor (Figure 7). Berm length had the largest independent effect (ΔAIC = +30.6; controlled $p < 0.001$), followed by flow accumulation (ΔAIC = +6.7; p = 0.003), soil texture (ΔAIC = +3.0; $p = 0.017$), and B-horizon presence (ΔAIC = +2.3; $p = 0.038$). B-horizon presence was marginally significant in univariate tests ($p = 0.059$), but significant in the controlled model ($p = 0.038$), indicating that its effect was partially masked by correlation with other predictors. Slope ($p = 0.920$) and landform ($p = 0.176$) were not significant after controlling for other predictors, indicating that the univariate signals were confounded by berm length, flow accumulation, and soil properties. The Random Forest model achieved a CV AUC of 0.71 ± 0.03 for condition (Figure 9). Permutation importance rankings were consistent with the LRT results: flow accumulation and berm length were the most important predictors, followed by slope and soil properties.

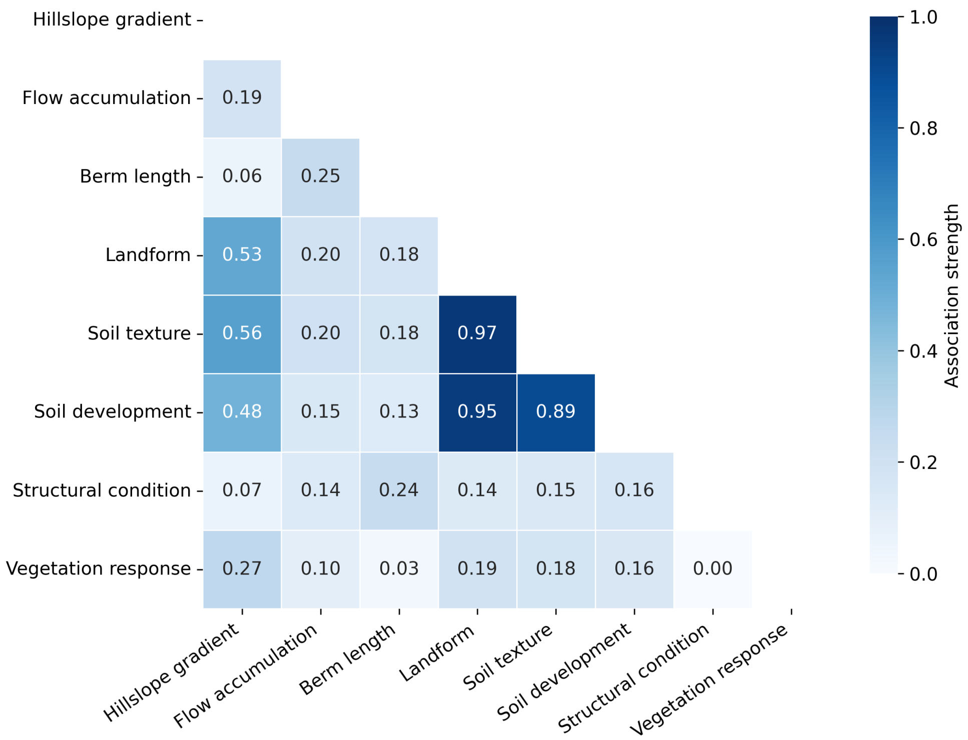

Figure 3: Pairwise association among the six candidate predictors and the two binary outcomes (berm structural condition and vegetation response), computed on the same predictor set and top-three-texture sample as the PCA (Figure 4; $n = 630$ berms). Because Spearman's ρ is appropriate only for ordered numeric variables, association is quantified with a measure matched to each pair type, all bounded on $[0, 1]$: |Spearman ρ| for numeric–numeric pairs, the Bergsma–Wicher bias-corrected Cramér's V for categorical–categorical pairs, and the correlation ratio η for mixed numeric–categorical pairs. The all-ones diagonal is masked. Landform, soil texture, and soil development are strongly inter-associated ($V = 0.89$–0.97), whereas both outcomes are only weakly associated with the predictors and with each other, motivating the controlled multivariate analyses below.

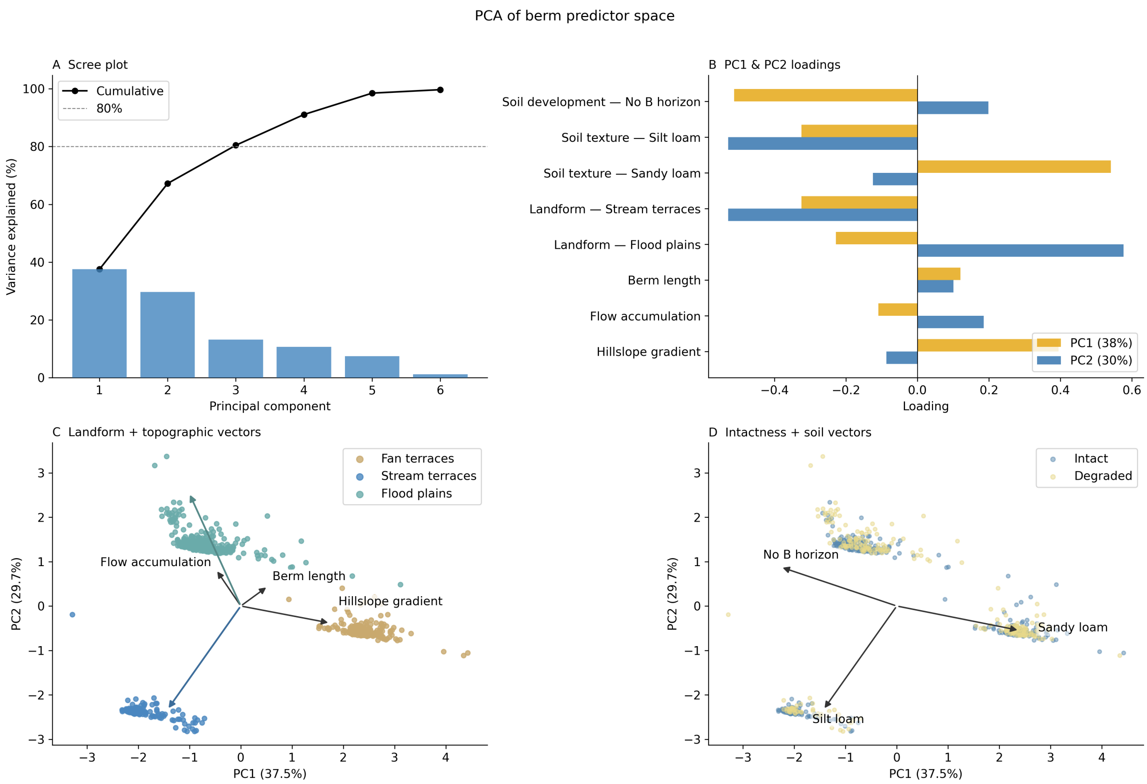

Figure 4: PCA of the berm predictor space. PC1 (38%) separates berms by soil texture and slope; PC2 (30%) captures landform variation. Landform clusters are well separated on PC2 but do not align with intactness, consistent with landform not being independently significant for condition.

3.3. Controls on vegetation response

Of 775 berms, 368 (47%) were classified as exhibiting a vegetation response ($\Delta S > 7\%$) and 407 (53%) were not. Univariate analyses showed that slope class, landform, soil texture, and B-horizon presence were all significantly associated with vegetation response, and berm length was not (Figure 6). Slope was the strongest predictor of vegetation response: berms on steep slopes (>2%) had significantly higher vegetation response rates than those on shallow slopes ($p < 0.001$). Among landforms, vegetation response rates differed significantly (χ^2 $p < 0.001$), with flood-plain berms showing the lowest rates of vegetation response. Soil texture was also highly significant (χ^2 $p < 0.001$), with sandy loam soils associated with greater vegetation response. B-horizon presence was associated with higher vegetation response ($p < 0.001$). Flow accumulation showed a weak univariate signal ($p = 0.031$), and berm length was not significant.

The controlled logistic regression (Figure 7) revealed that only slope (ΔAIC $= +18.8$; controlled $p < 0.001$) and soil texture (ΔAIC $= +5.6$; $p = 0.006$) retained significance after accounting for

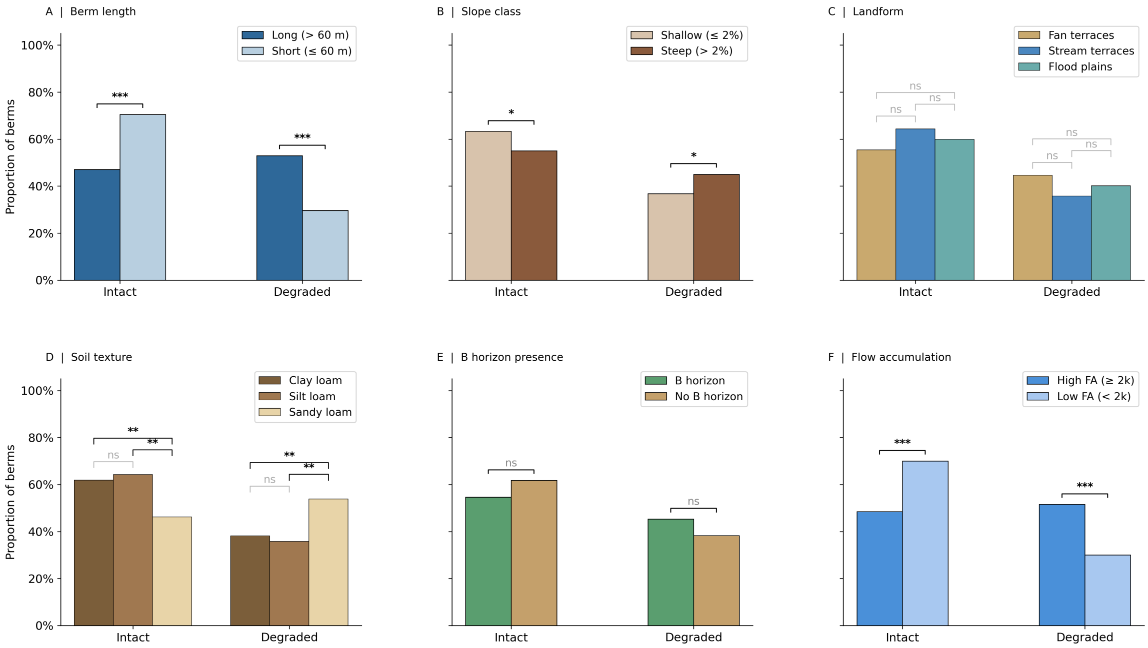

Figure 5: Six predictors of berm structural condition (Intact vs. Degraded). Two-category panels show two-sided Fisher's exact tests; multi-category panels show pairwise two-sided Fisher's exact tests with Benjamini–Hochberg FDR adjustment, with the overall χ^2 p-value reported in the body text. Berm length ($p < 0.001$), flow accumulation ($p < 0.001$), and soil texture (χ^2 $p < 0.001$) are the strongest predictors. Slope ($p = 0.016$) and B-horizon presence ($p = 0.059$) show weak-to-marginal univariate signals, though slope is not significant in the controlled analysis (Figure 7). Landform is not significantly associated with condition (χ^2 $p = 0.204$).

all other predictors. Landform is the clearest example of confounding effects of other variables: landform is not significant in the controlled model ($p = 0.310$), despite a strong univariate association ($p < 0.001$). Thus, the apparent effect of landform was explained by correlated differences in slope and soil properties, a result that is supported by the PCA analysis (Figure 4), in which landform clusters separate cleanly along the slope/texture principal components rather than along an independent landform axis. Presence of a B-horizon ($p = 0.112$), flow accumulation ($p = 0.108$), and berm length ($p = 0.712$) were also not significant after controlling for other variables.

The random forest model achieved a CV AUC of 0.64 ± 0.05 for vegetation response, with slope ranked as the most important feature by permutation importance, followed by flow accumulation and berm length (Figure 9). Predictive skill was lower for vegetation response than for berm condition; vegetation response integrates inter-annual climatic variability, time lags between berm construction and vegetation response, and biotic factors (e.g., grazing pressure, seed availability) that are not represented in the structural and edaphic predictors used here.

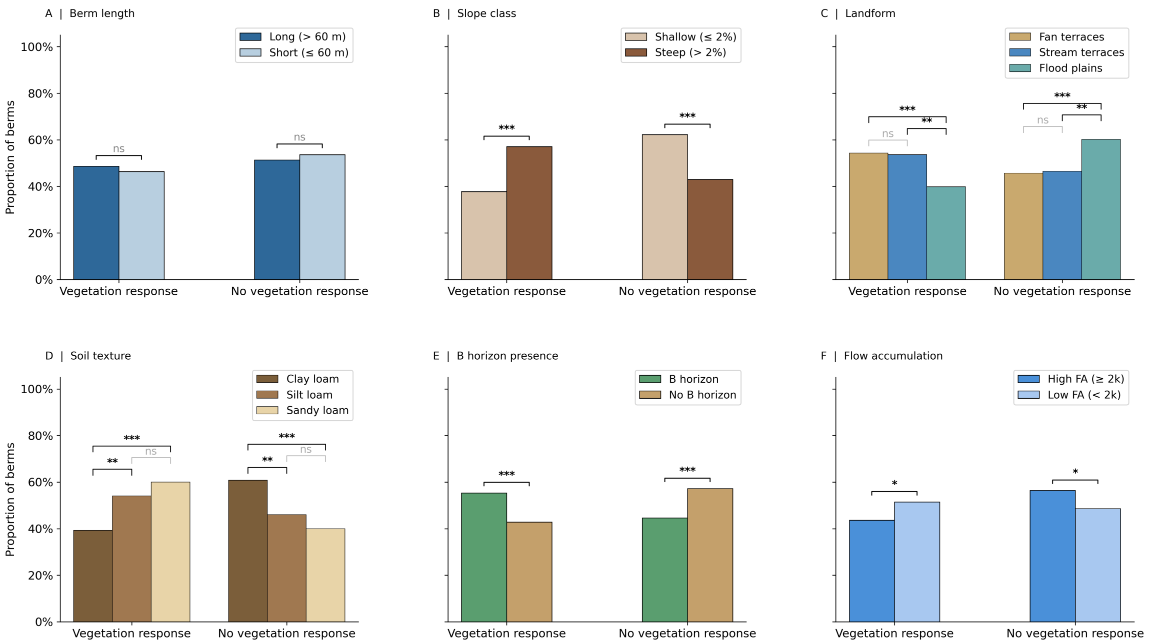

Figure 6: Six predictors of vegetation response. Two-category panels show two-sided Fisher's exact tests; multi-category panels show pairwise two-sided Fisher's exact tests with Benjamini–Hochberg FDR adjustment, with the overall χ^2 p-value reported in the body text. Vegetation response is significantly associated with slope class ($p < 0.001$), landform (χ^2 $p < 0.001$), soil texture (χ^2 $p < 0.001$), and B-horizon presence ($p < 0.001$), but not with berm length alone. However, the controlled analysis (Figure 7) shows that landform loses significance after accounting for slope and texture, indicating confounding.

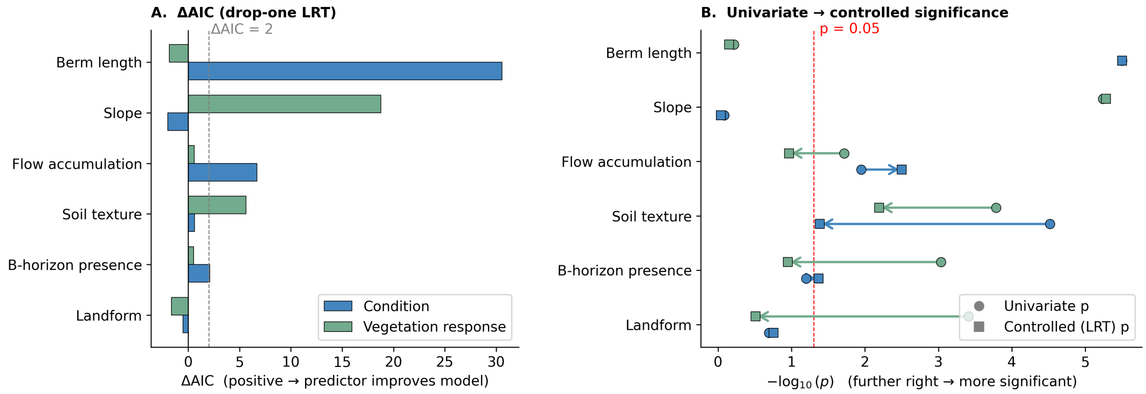

Figure 7: Independent predictor contributions to berm condition and vegetation response from controlled logistic regression. Panel A: ΔAIC from drop-one likelihood ratio tests. Panel B: shift from univariate to controlled p-values. Flow accumulation and berm length are the strongest controlled predictors of condition; slope dominates vegetation response. Several univariately significant predictors lose significance after controlling for confounders.

Independence of condition and vegetation response

Berm structural condition and vegetation response were statistically independent ($\varphi = -0.02$, $p = 0.612$; Figure 8), meaning that intact berms were not more likely to produce a vegetation response than degraded berms. When stratified by soil texture, only sandy loam berms showed a significant difference in vegetation response between intact and degraded berms (Figure 8).

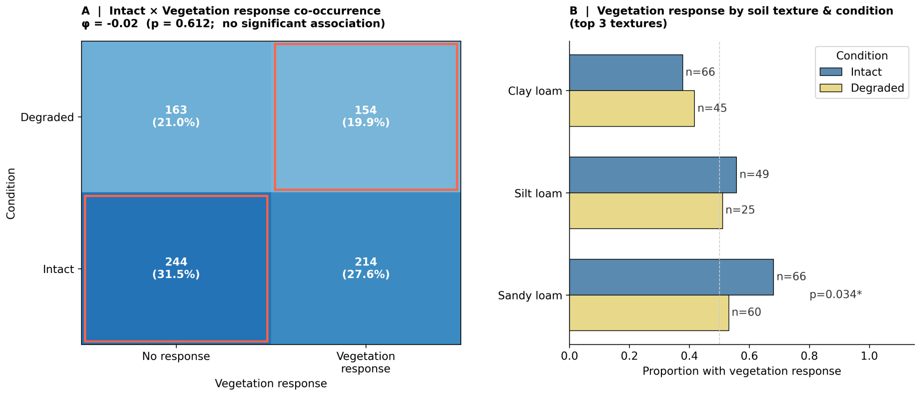

Figure 8: Berm structural condition and vegetation response are statistically independent overall ($\varphi \approx 0$); sandy loam shows the only significant difference in vegetation response between intact and degraded berms. Panel A shows the co-occurrence of condition and vegetation response across all berms; panel B stratifies the vegetation response by soil texture and condition.

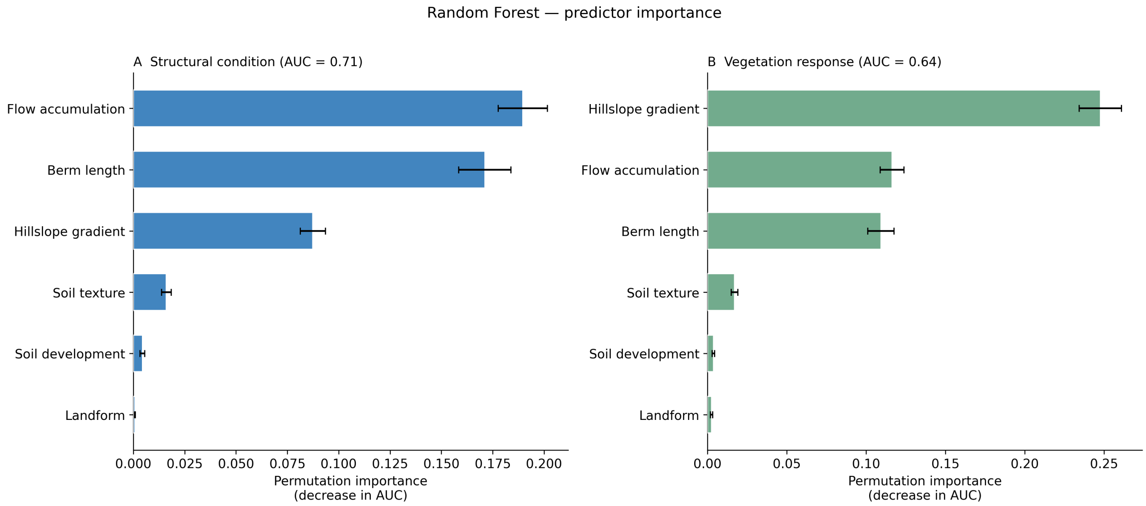

Figure 9: Random Forest permutation importance for predicting berm condition and vegetation response (CV AUC: condition = 0.71, vegetation response = 0.64). Flow accumulation and berm length are the best predictors of berm condition, followed by slope. For vegetation response, slope is the most important feature, followed by flow accumulation and berm length, with soil texture and soil development contributing less.

4. Discussion

Berm structural condition and vegetation response in the Altar Valley are shaped by largely distinct sets of environmental controls. Berm length, flow accumulation, and soil texture are the strongest predictors of structural condition, while slope and soil texture are the strongest predictors of vegetation response. Soil texture predicts both outcomes but in opposite directions: structural integrity is greater on finer textures, while vegetation response is greater on coarser textures. Landform, despite univariate associations with both outcomes, drops out of the controlled models, indicating that its apparent effect is mediated by correlated differences in slope and soil properties. Most

Table 2: Predictors of berm structural condition and vegetation response.

Predictor	**Condition**	**Vegetation response**
Berm length	Longer → more degraded; strongest predictor of condition	Not significant
Flow accumulation	Higher flow → more degraded	Not significant (controlled)
Soil texture	Finer textures → more intact	Coarse textures → more response; predictor of both outcomes but in opposite directions
Slope	Not significant	Strongest predictor of vegetation response; steeper → larger response
B-horizon presence	B horizon → more degraded; 'aquitard' effect confines lateral flow to the surface, driving flank failure	Not significant ($p = 0.112$); same aquitard effect promotes root-zone water
Landform	Not significant; confounded with slope and soil texture	Not significant; confounded with slope and soil texture

strikingly, the two outcomes are statistically independent at the berm-by-berm level ($\varphi = -0.02$, $p = 0.612$), suggesting that intact berms do not necessarily produce stronger vegetation responses than degraded ones. The remainder of this section develops the mechanistic interpretation of each finding. Table 2 summarizes main study findings.

4.1. Controls on structural condition

Berm length emerged as the single strongest predictor of structural condition (ΔAIC $= +30.6$), with longer berms showing significantly higher failure rates. Longer berms present more surface area for concentrated flow paths to develop, and their greater span increases vulnerability to localized weak points. Flow accumulation was the second-strongest predictor (ΔAIC $= +6.7$), indicating that hydrologic loading promotes berm failure independent of slope or landform. Soil texture also contributed independently (ΔAIC $= +3.0$), with berms on finer-textured soils (e.g., clay loam) more likely to remain intact, reflecting the greater cohesion and resistance to lateral scour of clay-rich materials [?]. B-horizon presence showed a suppression effect: its univariate signal was marginal, but it became significant after controlling for confounders, with B-horizon presence associated with *more* degraded berms. This pattern is mechanistically consistent with a near-surface aquitard interpretation, described below in Section 4.4. Interestingly, landform was not an independent predictor of structural condition ($p = 0.176$ after controlling for other variables). This result agrees with the PCA bi-plot (Figure 4), where landform clusters are well-separated by the first and second principal components but not aligned with intactness. This suggests that differences in berm condition among landforms may be attributable to correlated differences in soil properties and flow accumulation, rather than to landform per se. For example, flow accumulation is typically higher in flood plains than on fan terraces.

4.2. Controls on vegetation response

A different combination of predictors was significant for vegetation response: slope and soil texture, rather than berm length and flow accumulation. Steeper slopes were associated with greater vegetation response (ΔAIC $= +18.8$). A possible mechanism is that, on steeper terrain, berms intercept more runoff, increasing infiltration upslope and soil moisture available for vegetation. Soil texture also contributed independently (ΔAIC $= +5.6$), with coarser textures associated with greater vegetation response. This directional contrast is interpreted in the context of the inverse texture effect in Section 4.6. While landform showed a strong univariate association with vegetation response ($p < 0.001$), landform was not significant after controlling for slope and soil texture ($p = 0.310$). This suggests that landform acts as a proxy for underlying edaphic and topographic factors, rather than directly influencing vegetation response.

4.3. Varying roles of slope

Slope was the strongly predictive of vegetation response but not of structural condition. This divergent role may reflect that slope gradients in the Altar Valley are well below the thresholds at which slope-related mechanical failure is typically reported: 95% of berms occupy slopes $\leq$5% and the median is only $\sim$2%. For example, **?**] documented structural loss of check dams on slopes of 5–15% in semi-arid Spain, and **?**] recommended that water-harvesting structures avoid slopes exceeding $\sim$5% to limit erosive flow concentration. **?**] identified slope steepness as a primary control on check-dam failure in a global review of gully control structures. For the low gradients in the Altar Valley, increasing slope enhances the ecohydrologic impact of berms without independently affecting their structural integrity. That is, slope does not add predictive information for berm condition once other variables are accounted for (ΔAIC < 0). By contrast, even minor slope differences on gentle slopes may influence vegetation response: on steeper terrain, berms intercept and pond more runoff , increasing infiltration there.

4.4. Soil development

B-horizon presence may influence both outcomes through a vertical permeability contrast at the A–B horizon interface; a well-developed Bt horizon (illuvial clay accumulation) or Bk horizon (pedogenic carbonate accumulation) reduces saturated hydraulic conductivity at depth [**?** **?**]. For berm structural condition, this 'aquitard' effect may promote surface flows that concentrate water and erosive force at berm ends and cause flank failures. Berms on soils where a B horizon was present more frequently degraded; this effect strengthened after controlling for other covariates (see Figure 7; ΔAIC $= +2.3$, controlled $p = 0.038$). The same aquitard effect may support plant-available water by promoting soil moisture in the rooting zone after storms (see Figure 6). B-horizon presence was a strong univariate predictor of vegetation response ($p < 0.001$) but not significant ($p = 0.112$) once soil texture was included. Texture and B-horizon presence act on vegetation response in opposite directions: vegetation response was greater on coarser-textured soils (consistent with the inverse texture effect), while B-horizon presence augments rooting-zone storage, especially on otherwise free-draining soils. The lack of controlled significance of B-horizon presence may reflect this interaction: on the coarse-textured sites where the aquitard mechanism matters for plant water, soil texture and B-horizon are partially correlated, so the controlled model distributes the explanatory variance between them. Fine-textured soils could in principle degrade berm integrity through dispersive-clay failure mechanisms, in which sodic clays deflocculate in low-ionic-strength water and erode along internal seepage paths [**?**]; however, this mechanism is unlikely to be a significant factor in this dataset. Across all 775 berm locations, no soils are classified into Sodic or Natric subgroups, none are vertic, and only 0.3% (2 of 775) sit on smectitic-mineralogy components (SSURGO data). Sodium Adsorption Ratios are well below sodic thresholds (SAR > 13) at both the surface horizon and any horizon in the upper 1 m (95th-percentile $\mathrm{SAR}_{\mathrm{max,upper\ 1\,m}} = 1.0$).

4.5. Independence of condition and vegetation response

Berm structural condition was not significantly associated with vegetation response ($\varphi = -0.02$, $p = 0.612$; Figure 8). This finding is counterintuitive: intact berms that actively harvest runoff are expected to exhibit stronger upslope/downslope vegetation contrasts than degraded berms, which have presumably lost at least some of their hydrologic function.

One texture class is an exception. Among the three dominant textures, only sandy loam berms showed a significantly different vegetation response between intact and degraded berms, with a higher proportion of intact berms exhibiting vegetation response (52% vs. 38%, $p = 0.034$). This is consistent with the inverse-texture mechanism, under which berm-concentrated runoff produces detectable ecohydrologic differences specifically where coarser soils allow captured water to penetrate beyond the surface evaporation zone; sandy loam may represent the texture range where berm condition matters most for vegetation. The pattern may alternatively be a statistical artifact of multiple comparisons across texture classes.

Two non-exclusive mechanisms may reconcile this overall null result with expectation. First, the Altar Valley berm inventory does not resolve individual construction dates, and berm age is likely a confounding variable. Broader historical context suggests construction occurred within a ∼1935–1960 window [2]. Older berms have had more opportunity both to fail and to accumulate sediment, develop upslope soil structure, and establish perennial vegetation. Assuming berm age is positively correlated with both failure probability and vegetation response, two effects act in opposition. Berm age is a shared variable that links degradation to stronger vegetation contrasts, which alone would produce a positive association between the two. Loss of hydrologic function works in the reverse direction: degraded berms harvest runoff less effectively and so exhibit weaker vegetation response. These opposing effects may partially cancel, producing the observed near-zero association.

Resolving berm ages (e.g., through historical aerial photography or construction records) is a key next step for distinguishing this mechanism from the alternative below. Second, vegetation responses may persist after structural failure through legacy ecohydrologic modification. Sediment accumulation upslope of berms may produce locally thickened soil profiles with greater water-holding capacity. The resulting vegetation patch is expected to modify infiltration, organic matter inputs, and surface roughness, in ways that persist after the structure itself is breached or flanked.

Using a Saint-Venant solver, **?**] found that enhanced upslope vegetation can reproduce the infiltration enhancement of an intact berm under low-intensity storms, without the berm topographic structure, though the equivalence weakens for higher-intensity storms that vegetation captures less effectively than a physical wall. Under this hypothesis, enhanced vegetation may persist upslope of a degraded berm, not because it is presently functional, but because it once was.

These mechanisms are not mutually exclusive and likely operate in combination. Distinguishing them would require either age-resolved structural data or paired pre/post-failure observations, neither of which is currently available for the Altar Valley inventory. The independence finding should therefore be read as an empirical observation rather than as evidence against any specific mechanistic expectation.

4.6. Comparison with prior work

Soil texture and development have long been recognized as controls on earthen-structure stability and dryland vegetation response, but few studies have tested texture-dependence of both outcomes directly. For structural integrity, cohesion is the primary mechanism: fine-textured soils rich in clay and silt resist the lateral scour and particle detachment that initiate breach and flank failures, while coarse-textured soils with low cohesion are more susceptible to incision once concentrated flow develops [**? ?**]. In the Altar Valley specifically, water spreader berms fail by lateral scour (flanking) approximately twice as often as by breaching: of 667 spreader berms inventoried by Nichols and Degginger [1], 29% had been flanked while 15% had been breached. The texture-dependence of this failure pattern has not previously been examined; here we find that berms on finer-textured soils are more frequently intact, while berms on coarser-textured soils are more frequently degraded, consistent with cohesion-driven resistance to lateral scour.

For vegetation response, the inverse texture effect [**? ?**] predicts that runoff concentration should produce larger responses on coarser soils, where captured water penetrates below the evaporation zone, than on finer soils, where the same runoff may saturate only the upper profile and be lost to evaporation. The Altar Valley's 415 mm mean annual precipitation sits near the $\sim$370 mm crossover identified by **?**]; however, the valley's strongly monsoonal regime (>50% of rainfall in three months) and high evaporative demand reduce effective moisture availability relative to comparably-watered Great Plains sites, and berm-concentrated runoff produces pulsed wetting events for which the inverse-texture logic applies more directly than to background precipitation.

Our findings are consistent with **?**], who classified grass, shrub, and bare-soil cover from 0.75 m RGB imagery upslope and downslope of 181 Altar Valley berms stratified into three soil texture groups (clay loam, sandy loam, loamy sand). The study found that upslope-vs-downslope grass cover contrasts were absent on clay loam soils across all berm conditions, but were significant for intact and flanked berms on the coarser sandy loam and loamy sand soils. On loamy sand soils, the grass cover contrast disappeared at breached berms, suggesting that re-establishing hydrologic connectivity at coarse-soil sites may dissipate the vegetation response that berms created. Our finding that vegetation response is highest on sandy loam berms extends this pattern to a larger inventory.

4.7. Management implications

Because structural condition and vegetation response respond to different predictors and are statistically independent at the berm-by-berm level, management strategies should target each outcome separately. Shorter berms on finer-textured soils with lower flow accumulation are more likely to remain intact; vegetation responses are larger on steeper slopes and coarser-textured (sandy loam) soils.

The independence finding has management implications: maintenance prioritized solely by structural condition may not target the berms providing the greatest ecological benefit, and decisions about which berms to repair, remove, or leave in place should weigh both dimensions.

5. Conclusion

Structural condition and vegetation response of water spreader berms in the Altar Valley are shaped by distinct environmental controls. Structural condition is primarily determined by berm length, flow accumulation, and soil texture, whereas vegetation response is more sensitive to slope and soil texture. Landform, despite strong univariate associations, is confounded with these underlying controls and does not independently predict either outcome once other variables are accounted for. Structural condition and vegetation response are statistically independent, highlighting the need to evaluate both physical integrity and ecological function separately.

The predictive accuracy of our models is modest (CV AUC = 0.71 for condition, 0.64 for vegetation response). Semi-arid rangelands integrate soil heterogeneity, storm-by-storm hydrologic variability, vegetation dynamics, and the ecohydrologic feedbacks among them, most of which operate at scales finer than our SSURGO- and DEM-derived predictors can resolve. Our analysis is correspondingly coarse: we identify broad landscape-scale controls rather than mechanistic predictors of individual berm outcomes.

For practitioners, the controls identified here, such berm length, flow accumulation, soil texture, slope, and soil development, can be assessed from existing terrain and soil survey data, providing a basis for prioritizing maintenance or removal across large berm inventories without site-by-site field assessment.

The use of satellite vegetation indices such as SAVI offers a scalable approach for monitoring berm impact on rangeland vegetation over time. A landscape-informed approach to earthwork design and management that accounts for soil-geomorphic controls can improve both water retention and vegetation outcomes in dryland ecosystems. Open questions remain about the role of berm age, maintenance history, and upstream land-use change in modulating these outcomes; addressing them will require longitudinal monitoring and spatially explicit hydrologic modeling.

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